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GNIOT Institute of Management Studies – PGDM (Batch 2022-24)**(TRIMESTER – II) END TERM EXAMINATION, DECEMBER - 2022****INFORMATION SYSTEMS MANAGEMENT FOR BUSINESS**

[Time: 2 Hours 30 Minutes]

[Maximum Marks: 100]

INSTRUCTIONS:

- (i) There are **THREE** sections in this question paper.
- (ii) Attempt questions from each section as per the directions.
- (iii) Make suitable assumptions wherever necessary.

SECTION-A**(Short Answer Type Questions, do not exceed 100 words)****Q.1. Write short notes on the following:****(10×3=30)**

- (a) Data Vs Information
- (b) OLTP and OLAP
- (c) Role and responsibilities of Middle layer.
- (d) Data Mart
- (e) Planning and strategy in reference decision making.
- (f) Pull Vs. Push based supply chain model.
- (g) Characteristics of Data Warehouse.
- (h) With the help of suitable diagram explain E-R diagram.
- (i) Positive and negative impact of Information Systems.
- (j) Group Decision Support Systems (GDSS)

SECTION-B**(Long Answer Type Questions)****Attempt all questions of the following:****(4×10=40)**

Q.2. (a) Assume an old man visits your organization and asks you to show him the MIS of your organization. How would you make him understand the structure of MIS?

OR

(b) “A Management Information System is an information system used for decision-making.” Justify your answer with suitable statement.

Q.3. (a) Distinguished Data Warehouse and Database? Draw the diagram of Data Warehousing Architectures and explain each of its components?

OR

(b) “Data mining is a process of finding KDD from the huge amount of data.” Critically evaluate the sentence with the help of suitable diagram.

Q.4. (a) Explain all the seven phases of software development life cycle models?

OR

(b) Mention trends in the global Business environment that has made information systems so important. Also explain which IT technology affected that type and nature of organisations and how?

Q.5. (a). Explain the Enterprise Resource Planning System? Also discuss the concepts and challenges of ERP.

OR

(b) “Information Technology is the enabler of supply chain management system.” Comment on this in the light of how SCM has been benefitted by IT tools.

SECTION-C

Q.6. Attempt all parts of the following:

(2×15=30)

With the advent of science and technology, technological innovations have become a pre-requisite to achieve operational and strategic excellence. Organizations by using state-of-the-art systems, are striving for their best to achieve sustainable and long-term competitive advantage. It has become sine qua non to use modern tools for the effective achievement of goals and objectives. Its importance can also be observed from the increased spending on Research & Development in order to pace up with the today's dynamic business environment. Wal-Mart, founded by Sam Walton in 1962, is the world's largest retailer that employs about 2.1 million associates worldwide, in more than 8,400 stores, including 8, discount stores, 3,100 combination discount and grocery stores.

Wal-Mart is serving its customers and members more than 200 million times per week at more than 8,613 retail units under 55 different banners in 15 countries. Wal-Mart has ranked first among retailers in Fortune Magazine's 2010 Most Admired Companies survey with fiscal year 2010 sales of \$405 billion. Sam Walton's belief was to build an empire by providing value to its customers and empowering employees, also known as associates. The way the Wal-Mart operates is so fascinating and impressive that Jack Welch, CEO of General Electric once said: "Many of our management teams spent time there observing the speed, the bias for action, the utter customer fixation that drives Wal-Mart." The use of high-tech Information technology has always been an essential ingredient for Wal-Mart's growth. Since its inception, it has used the IT systems well enough for the Inventory, administrative, customers and suppliers' management.

Wal-Mart's investment in technology started with the computerized accounting systems and since then the continuous adoption of latest technologies has made Wal-Mart a market leader in the whole retail industry. Wal-Mart developed a well-equipped computerized Point of Sale (POS) system capable of identifying each unit sold, creation of accurate sales receipt and storing all the information item-by-item for sales analysis and inventory reordering purposes. It helped company to avoid overstocking of merchandise through effective information usage and also assisted them in investing in only those merchandise that were profitable. It has also helped them figure out best-selling products and prospective products that customers want. Analysts say that it is Wal-Mart's Point of Sale system that has helped them achieve the world's largest market share and revenues in retail industry. Wal-Mart, in 2004, was the first in retail industry to adopt electronic radio frequency identification (RFID) tags for inventory distribution management. RFID technology, which is still being used, helps them keep track of inventory movement across its supply chain. Wal-Mart was trailing K-Mart in 1980's but with the adoption of latest technologies, it has now become the trend setter in retail industry all over the world. Wal-Mart is currently working on a project to make best out of wireless technology. In near future, they have plans to develop a remote system that would allow customers to just walk into their store and use their own devices (Phones or PDA's) to get whatever they want. i.e., they might go online to their website, while at the store, and find things for themselves or just compare the products' prices etc.

Customer service, best shopping environment and experience, one-stop shopping, best value products, customer convenience of online shopping, good customer relations are the blessings that Wal-Mart provides using cutting-edge technology. Latest wireless system at their stores, online web-retailing, vast width and hierarchy of product items, and stores at multiple remote sites are the benefits customers of Wal-Mart cherish. Wal-Mart is considered to be a trend setter in retail industry and it's all due to its inclination towards achieving impossible through modern technology. For example, Wal-Mart once tried to develop three experimental stores in McKinney, Aurora and Las Vegas that were equipped with latest technologies or concepts like wind turbines, photovoltaic solar panels, bio-fuel-capable boilers and water-cooled refrigerators. Irrespective of the criticism it had on environmental issues, Wal-Mart's initiative was considered to be a positive step towards change and innovation and it provided them with an opportunity to become the biggest seller of organic milk and the biggest buyer of organic cotton in the world, as well as reducing packaging and energy costs. It has been quoted in 2006's annual report of Wal-Mart that in order to compete against upscale and appealing store, such as 'Target', they have launched a new Supercentre concept in Plano, Texas. The new store had wood floors, wider aisles, a sushi bar, a coffee shop with free Wi-Fi Internet access, and more expensive beers, wines and other electronics goods.

By using advance forecasting techniques and integrating different technologies, Wal-Mart is planning to deploy simulation software that would help them simulate business events that are yet to happen.

Wal-Mart has got a strategic plan of introducing a concept of self-service. Self-service technology, aimed at eliminating paper and paper forms, will be provided to their associates, prospective associates, customers and members. It was said by Kevin Turner, CEO of Wal-Mart's subsidiary corporation, in his interview with 'CIO' personnel that "having an associate portal and devices on our sales floors will let customers and members get product information, and let us do computer assisted selling". Wal-Mart has been outstanding in its investments in Information technology from all prospects.

IT systems at Wal-Mart allow them to reduce transactional as well as operational costs at their distribution centres and it has also helped them to support Wal-Mart's long-term strategy of owning the distribution centres and maintaining long-term relationships with their suppliers. Also, improvement in productivity due to IT systems is dramatic at Wal-Mart. It has helped them offer products at such a nominal cost that a small-town merchant is unable to provide and this is remarkable achievement of Wal-Mart. It has become a sustainable competitive advantage for them over their competitors and a reason of their growth and profitability. We are aware of a fact that every benefit is accompanied with some problem but Wal-Mart has successfully defied its weaknesses and have used its strength well to take advantage from the prospective opportunities and avoiding possible threats

6(a) How IT and IS may be used to help the company operate and achieve its objective. Also discuss the challenges faced by the CIO - which are the most important and why?

6(b) What role does information technology play in the business? How is it helping Walmart refine its business strategies?